



Two-stream convolutional LSTM for precipitation nowcasting

Suting Chen^{1,2} · Xin Xu¹ · Yanyan Zhang¹ · Dongwei Shao¹ · Song Zhang¹ · Mingjian Zeng³

Received: 28 June 2021 / Accepted: 16 December 2021 / Published online: 6 January 2022
© The Author(s), under exclusive licence to Springer-Verlag London Ltd., part of Springer Nature 2022

Abstract

Reliable precipitation nowcasting is essential to many fields, which can guide people to reasonably carry out production activities and respond to rainstorm disasters. However, precipitation nowcasting is a very challenging task because of correlation and heterogeneity both in space and in time. Most previous studies have not adequately captured the long-term and long-range spatiotemporal dependencies in the data, leading to insufficient modeling and poor prediction performance. To make more accurate prediction, we propose a novel deep learning model for precipitation nowcasting, called two-stream convolutional LSTM which includes short-term sub-network and long-term sub-network. The two sub-networks, respectively, make predictions on inputs at different time intervals to capture the heterogeneity of rainfall data. On this basis, an innovative recombination module is proposed to fuse the outputs of two sub-networks. In addition, we embed the 3D convolutions and self-attention mechanism to construct a new memory cell, named 3D-SA-LSTM, to extract the spatiotemporal feature. Two-stream convolutional LSTM achieves the state-of-the-art prediction performance on a real-world large-scale dataset and is a more flexible framework that can be conveniently applied to other similarly time series prediction tasks: traffic forecasting and planning, financial analysis and management, actions recognition and prediction, etc.

Keywords Precipitation nowcasting · Two-stream convolutional LSTM · 3D convolutions · Self-attention mechanism

1 Introduction

Precipitation nowcasting has strong forecasts timeliness, commonly from 1 to 6 h, and is aimed at the severe convective weather system with short duration and strong suddenness. It is difficult to predict because of the complex mechanism of the system, the small scale of spatiotemporal, and the fast generation and dissipation speed [1, 2]. The conventional technique of precipitation nowcasting mainly includes statistical precipitation forecast method, precipitation forecast method based on the numerical model and precipitation forecast method based on radar extrapolation

[3–6]. But the above approaches all give specific models, which means these approaches are difficult to adequately mine the relationship between data, directly resulting in the low accuracy of forecasting. With the development of monitoring technology, a large amount of meteorological data has been collected, and deep learning has shown huge potential in this field [7].

From the perspective of deep learning, precipitation nowcasting can be regarded as a time series prediction problem; this problem has long been a subject that has attracted researchers from a diverse range of fields [8–16]. Recurrent neural network (RNN) and long short-term memory (LSTM) are often used to solve sequence problems and have achieved great results [17–20]. Ranzato et al. proposed a model by learning to predict missing frames or extrapolate future frames from an input video sequence [21]. Srivastava et al. first introduced LSTM to video analysis and extended sequence-to-sequence model to learning video representation [20]. Finn et al. developed an action-conditioned video predict model that can predict previously unseen objects [22]. Kalchbrenner et al. proposed video pixel networks (VPN) which encodes the time, space and color structures of videos as a four-dimensional

✉ Suting Chen
sutingchen@nuist.edu.cn

¹ Jiangsu Collaborative Innovation Center of Atmospheric Environment and Equipment Technology(CICAEET), Nanjing University of Information Science & Technology, Nanjing 210044, China

² Nanjing University of Information Science & Technology, Wuxi Institute of Technology, Wuxi 214100, China

³ Jiangsu Institute of Meteorological Sciences, NanJing 210009, China

dependency chain. It achieves sharp prediction results but suffers from a high computational complexity [23]. [24, 25] introduce generative adversarial network (GAN) [26] used for future video prediction; both papers illustrate that GAN is helpful for reducing the blur of predictions. Compared to ordinary convolutional neural networks (CNNs), 3D convolutional neural networks (3D-CNNs) and self-attention mechanism have been shown to help improve the ability to learn the features [27–31].

Klein et al. [32] proposed the dynamic convolutional layer and used it for weather forecasting. Shi et al. constructed the convolutional long short-term memory (ConvLSTM) to extract spatiotemporal features more effectively by integrating CNNs into LSTM; the approach gives more accurate prediction performance than real-time optical flow by variational methods for echoes of radar (ROVER) algorithm [8, 33]. Furthermore, a more advanced trajectory gated recurrent unit (TrajGRU) that can actively learn the location-variant structure for recurrent connections is proposed [34]. Besides, many researchers have also proposed many variants of ConvLSTM, e.g., predictive recurrent neural networks (PredRNN) [35], Eidetic 3D memory (E3D-LSTM) [36], memory in memory (MIM) [37], and physical cell (PhyCell) [38]. In Table 1, we summarize the methodology of the relevant works just presented.

However, most of the previous work has some deficiencies. First, these methods do not take into account the data heterogeneity between time periods. Figure 1 displays the changes of the rainfall process in two different time intervals. In the first row, rainfall distribution has only a small change in area and value. On the contrary, the area and value of rainfall in a longtime interval have obvious changes, and there is a clear movement trend from left to right. If data heterogeneity cannot be captured, it is difficult to make accurate predictions. Second, the ConvLSTM and its variants learn spatial features by using a fixed convolution kernel and stacked RNN layers. Its real receptive field is still smaller than the theoretical receptive field and cannot effectively learn long-range spatial dependencies; other approaches are insufficient in learning long-term time dependencies [39].

Therefore, we aim to solve the aforementioned problems by proposing a novel model and a memory cell. Inspired by two-stream ConvNet and squeeze-and-excitation (SE) blocks, we investigate a model that includes short-term stream and long-term stream, which are then combined with their outputs through the following recombination module [40, 41]. The short-term stream performs precipitation nowcasting features from recent data, while the long-term is trained to precipitation nowcasting from periodic data; two sub-networks are combined to capture the heterogeneity of rainfall data. For the new memory cell, we

propose the 3D-SA-LSTM which combined 3D convolutions (3D-Conv) and self-attention mechanism; 3D-Conv can improve the learning ability of local motion perception and recall short-term historical memory, and the introduction of self-attention mechanism helps memory cells capture the long-range spatial dependencies. Our two-stream convolutional LSTM achieves the state-of-the-art results on a real-world large-scale rainfall dataset for precipitation nowcasting.

2 Materials and methods

The rainfall dataset contains per minute rainfall data from June to October in 2018 and 2019, which cover most areas of North China. These data were collected by ground weather stations; compared with using the Z–R relationship to convert the radar echo data into the rainfall intensity value, the ground weather station can directly get the rainfall intensity value, which can avoid introducing unnecessary system errors. Since rainfall events rarely occur, we select some months with more rainfall to form our final dataset and use the rainfall data in 2019 as the training set, the data in June 2018 as the validation set, and the data in July and August 2018 as the test set. The proportion of different rainfall levels in dataset is shown in Table 2. To save computing resources, we draw the rainfall map as a 64×64 grayscale image. When the rainfall intensity value in a place is 1 mm/h, the grayscale value of the pixel at the corresponding position in the image is 6.

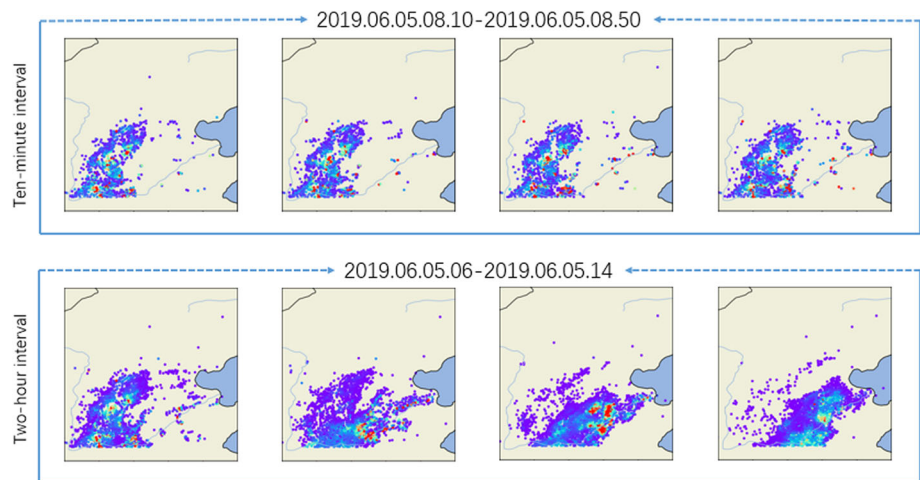
To more effectively capture the heterogeneity of the above-mentioned different time interval data, we design a deep learning-based precipitation nowcasting model, named two-stream convolutional LSTM, to predict future rainfall data. Figure 2 demonstrates the architecture of two-stream convolutional LSTM that consists of two separate sub-networks and a recombination module; the two sub-networks are long-term sub-network and short-term sub-network, respectively. The short-term sub-network is used to capture spatiotemporal relationships in recent times, i.e., the temporal and spatial motion in local regions. In contrast, the long-term sub-network is better at learning some higher-level features, i.e., the overall movement trend during rainfall. The recombination module is designed to fuse the outputs that form the two sub-networks; compared to directly concatenating the feature map or assigning weights to the feature map, the module can better increase the representational ability of the features by nonlinearities fusion.

The structure of sub-network is introduced in Sect. 2.1. And then we discuss how to integrate the 3D-Conv and self-attention mechanism into the memory cells, which is

Table 1 Summary of the existing works for time series prediction

References	Methodology
[8]	Integrated the convolution operator into recurrent state transition functions
[34]	Provided a benchmark for precipitation nowcasting and proposed a new TrajGRU model with the ability of learning the recurrent connection structure
[35]	Proposed a new memory cell: Spatiotemporal LSTM (ST-LSTM) and increased the direction of memory flow in the network
[36]	Integrated 3D-Convs into state transitions to perceive short-term motions and designed a memory attentive module controlled by recurrent gates
[37]	Proposed a new recurrent neural network to enable nonstationary modeling in the spacetime contexts
[38]	Proposed PhyDNet for disentangling prior dynamical knowledge from other factors of variation and proposed physical cell (PhyCell) for performing PDE-constrained (partial differential equations) prediction in latent space

Fig. 1 Rainfall map in two different time intervals. First row: each rainfall map is drawn from rainfall data within 10 min. Second row: each rainfall map is drawn from rainfall data within 2 h



detailed in Sect. 2.2. Finally, we display the recombination module in Sect. 2.3.

2.1 Encoding-forecasting structure

In our network, two sub-networks have some differences in the channel of the input and the convolution kernel size, and the rest are exactly the same. Take short-term sub-network as an example to illustrate the structure of the sub-network.

As shown in Fig. 3, the sub-network is composed of two independent parts: the encoding network and the

forecasting network, which can be classified into a broader structure: encoding-forecasting structure proposed by [8]. Based on this structure, we added an extra zigzag direction information flow that sends the state of the top layer memory cell at the previous timestamp to the bottom layer memory cell at the next timestamp, which helps to learn spatiotemporal dependencies by increasing the feature interactions at different-level convolutional layers. The high-level states have richer spatiotemporal relationships, which can guide low-level cells to update their states and further improve the accuracy of prediction. We assume that observation at time t is $X_t \in R^{M \times N}$ and the spatiotemporal sequence $\langle X_1, X_2, \dots \rangle \in R^{S \times M \times N}$ contains multi-frame continuous rainfall maps; the dimension S is temporal depth, while the dimension M and the dimension N are height and width of the rainfall maps, respectively. At time t , the encoding-forecasting structure predicts the most probable future K -steps sequence: $\langle \tilde{X}_{t+1}, \dots, \tilde{X}_{t+K} \rangle$ based on previous J -steps observations which include the current one:

Table 2 Rain rate statistics in the rainfall dataset

Rain rate (mm/h)	Rainfall level	Proportion (%)
$0 \leq x \leq 0.1$	No	87.00
$0.1 \leq x \leq 1$	Light	9.26
$1 \leq x \leq 10$	Moderate	3.13
$10 \leq x \leq 20$	Heavy	0.34
$20 \leq x$	Rainstorm	0.25

Fig. 2 Two-stream convolutional LSTM architecture. The yellow arrows indicate the direction of information flow. The figure is simplified for illustration, and the specific details will be explained in later sections

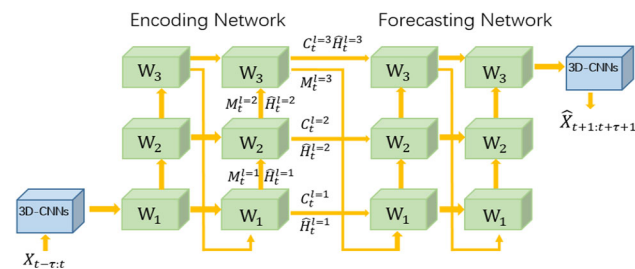
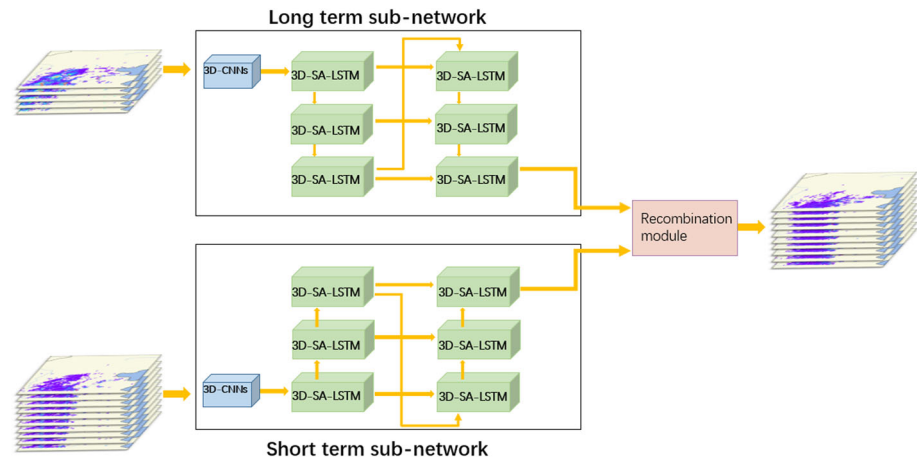


Fig. 3 Structure of short-term sub-network. W_n represents the 3D-SA-LSTM cells; $\hat{H}_t^l$, C_t^l and M_t^l are the three states of 3D-SA-LSTM cells at timestamp t , layer l

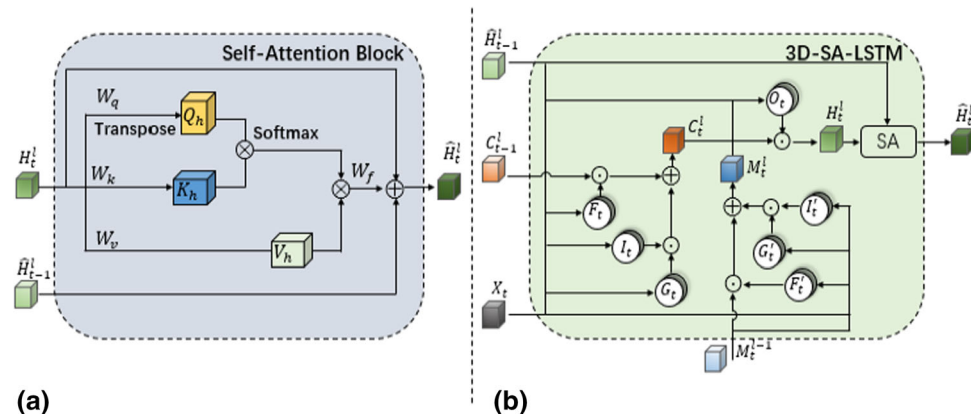
$$\begin{aligned} \tilde{X}_{t+1}, \dots, \tilde{X}_{t+K} &= \arg \max_{X_{t+1}, \dots, X_{t+K}} P(X_{t+1}, \dots, X_{t+K} | X_{t-J+1}, \dots, X_t) \\ &\approx \arg \max_{X_{t+1}, \dots, X_{t+K}} P(X_{t+1}, \dots, X_{t+K} | f_{\text{encoding}}(X_{t-J+1}, \dots, X_t)) \\ &\approx g_{\text{forecasting}}(f_{\text{encoding}}(X_{t-J+1}, \dots, X_t)) \end{aligned} \quad (1)$$

It is worth noting that the 3D-CNNs of forecasting network are integrated into the recombination module.

2.2 3D-SA-LSTM

Self-attention mechanism and 3D-CNNs have been proven to help improve the performance of time series prediction problems. Therefore, we design a self-attention (SA) block (c.f. Fig. 4a) and combined it with 3D-Conv to construct a new memory cell: 3D-SA-LSTM (c.f. Fig. 4b). We use 3D-Conv instead of the ordinary convolutions as a feature extractor, which can simultaneously capture spatiotemporal dependencies between adjacent frames to make 3D-SA-LSTM more adept at learning and describing local or short-term motions. The main equations of the 3D-SA-LSTM are given as follows:

Fig. 4 Self-attention block a and b 3D-SA-LSTM. The cubes represent higher-dimensional hidden states and memory states. The cylinder represents higher-dimensional gates



$$\begin{aligned}
I_t &= \sigma(W_{xi} * X_t + W_{hi} * \hat{H}_{t-1}^l + b_i) \\
F_t &= \sigma(W_{xf} * X_t + W_{hf} * \hat{H}_{t-1}^l + b_f) \\
G_t &= \sigma(W_{xg} * X_t + W_{hg} * \hat{H}_{t-1}^l + b_g) \\
C_t^l &= F_t \odot C_{t-1}^l + I_t \odot G_t \\
I_t' &= \sigma(W_{xi}' * X_t + W_{mi}' * M_{t-1}^{l-1} + b_i') \\
F_t' &= \sigma(W_{xf}' * X_t + W_{mf}' * M_{t-1}^{l-1} + b_f') \\
G_t' &= \sigma(W_{xg}' * X_t + W_{mg}' * M_{t-1}^{l-1} + b_g') \\
M_t^l &= F_t' \odot M_{t-1}^{l-1} + I_t' \odot G_t' \\
O_t &= \sigma(W_{xo} * X_t \\
&\quad + W_{ho} * \hat{H}_{t-1}^l + W_{co} * C_t^l + W_{mo} * M_t^l + b_o) \\
H_t^l &= O_t \odot \tanh(W_{1 \times 1 \times 1} * [C_t^l, M_t^l]) \\
\hat{H}_t^l &= SA(H_t^l, \hat{H}_{t-1}^l)
\end{aligned} \quad (2)$$

where σ is the sigmoid activation function, $*$ and $\odot$ are the 3D-Conv operations and Hadamard product, respectively, $W_{1 \times 1 \times 1}$ is the $1 \times 1 \times 1$ convolutions used to convert the channel number of hidden states, SA is the self-attention module, and $[\cdot, \cdot]$ denotes concatenation operation. 3D-SA-LSTM includes four inputs: X_t is the cell input obtained by encoding the input sequence by 3D-Conv; $\hat{H}_{t-1}^l$, the hidden states from previous timestamp; C_{t-1}^l , the standard temporal memory states from previous timestamp; and M_{t-1}^{l-1} , the spatiotemporal memory is vertically transmitted to the current cell from the $l-1$ layer at the same timestamp, for $l=1$, $M_t^{l-1} = M_{t-1}^l$. The initial state of memory cells in the first timestamp of the coding network is set to 0; the input tensor of memory cells in forecasting network is also set to 0.

To learn the long-range spatial dependencies, we improved the non-local block [29] and proposed a new SA block, the output of SA block: $\hat{H}_t^l$ is generated by mapping the original hidden state H_t^l to other feature spaces and adding the previous hidden state $\hat{H}_{t-1}^l$. The formulations for mapping the original hidden states H_t^l to different feature spaces are as follows:

$$\begin{aligned}
Q_h &= W_q \otimes H_t^l \in R^{\hat{c} \times T \times H \times W} \\
K_h &= W_k \otimes H_t^l \in R^{\hat{c} \times T \times H \times W} \\
V_h &= W_v \otimes H_t^l \in R^{\hat{c} \times T \times H \times W}
\end{aligned} \quad (3)$$

where $\otimes$ is the matrix multiplication after reshaping matrices into two-dimensional forms, $\hat{c}$ is the number of channels, and $\{W_q, W_k, W_v\}$ is a set of weights for $1 \times 1 \times 1$ convolutions. The query Q_h is the attention weight of spacetime position; the key K_h is the weight index that enumerates all possible positions; the value V_h represents the value of feature maps. The similarity scores of each pair of points e_{ij} can be calculated by matrix multiplication:

$$\begin{aligned}
f(Q_h, K_h) &= Q_h^T K_h \in R^{T \times H \times W \times T \times H \times W} \\
e_{ij} &= (H_i^T W_q^T)(W_k H_j^l), i, j \in \{1, 2, \dots, N\}
\end{aligned} \quad (4)$$

where i, j is the index of position and where $N = T \times H \times W$. The similarity scores are normalized along columns:

$$\alpha_{ij} = \frac{\exp e_{ij}}{\sum_{k=1}^N \exp e_{ij}} \quad (5)$$

where y is the output signal of the same size as input and the aggregate feature of the i th position is calculated with a weighted sum across all positions:

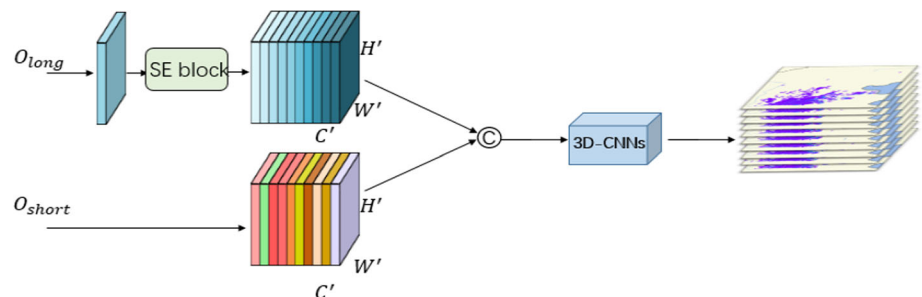
$$y_i = \sum_{k=1}^N \exp e_{ij}(W_v, H_{t_j}^l) \quad (6)$$

Finally, the output feature $\hat{H}_t^l$ can be formulated as follows:

$$\begin{aligned}
\hat{H}_t^l &= SA(H_t^l, \hat{H}_{t-1}^l) \\
&= W_f y + \beta H_t^l + (1 - \beta) \hat{H}_{t-1}^l
\end{aligned} \quad (7)$$

where '+' denotes a residual connection [42] and β is the hyper-parameter that controls how much historical memory states will be retained. The residual connections allow SA block to be embedded in other LSTM cells without destroying the original output, such as W_f is initialized as 0 and β is initialized as one, and the SA block output will be

Fig. 5 The recombination module. O_{long} and O_{short} are the output of the long-term sub-network and the short-term sub-network, respectively. SE block is the squeeze-and-excitation block. H' , W' and C' are the feature maps dimension. $\odot$ is the channel concatenation



equal to the input. Benefitting from the introduction of 3D-Conv and self-attention mechanism, 3D-SA-LSTM's ability to recall long-term memory and long-range dependencies surpasses ConvLSTM and its variants.

2.3 Recombination module

As shown in Fig. 1, two rainfall maps drawn at different time intervals have obvious differences in data distribution. If we fail to identify and quantify the role of long-term sub-network in the prediction of different time lengths, it will directly lead to insufficient modeling of the correlation of rainfall data and poor prediction performance. Therefore, we present a recombination module (see Fig. 5) to fuse the output of two sub-networks, which can appropriately quantify the contribution of long-term sub-network for each temporal dimension.

Due to the output of the long-term sub-network and the short-term sub-network having some differences in channel dimensions, the recombination module introduces the SE block to adjust the output channels' number of the former to make it the same as the output of the latter [40]. The SE block allows the model to pay more attention to the channel features with the most information and suppress the unimportant channel features, thereby boosting the representational power of feature map. And then, we perform feature map fusion through channel concatenate operation, and finally, 3D-CNN decodes the obtained feature map to give the final prediction.

3 Experiments and discussion

In this section, we evaluate the proposed two-stream convolutional LSTM model using a real-world large-scale rainfall dataset for precipitation nowcasting. The details of evaluation indicators are given in Sect. 3.2. Then, we show the results of the ablation study and comparisons with other existing methods in Sects. 3.3 and 3.4, respectively.

3.1 Implementation details

We compare our two-stream convolutional LSTM with the following three baseline methods:

- ConvLSTM: Convolutional LSTM [8] is the pioneering work of deep learning in the field of precipitation nowcasting.
- TrajGRU: Trajectory gated recurrent unit [34] has been applied in real-world precipitation nowcasting task.
- MIM: Memory in memory [37] achieves the current state-of-the-art performance on a series of spatiotemporal prediction tasks.

Our model has four layers in all experiments like previous work to achieve fair comparison. We run all the experiments on a computer with a single NVIDIA GTX TITAN-X GPU. All models are using the ADAM optimizer [43] with an initial learning rate of 0.001, and the batch size is set to 4, and stop training after 100,000 iterations. Additionally, LayerNorm [44] is also used in the training process. In our model, the input sequence of the short-term sub-network includes 10 frames of images, and each frame represents a 6-min rainfall map; the input sequence of the long-term sub-network includes 5 frames of images, and each frame represents a 1-h rainfall map. It is worth noting that the last moment of the two input sequences is the same moment. The input of the rest of the models is the same as the short-term sub-network. We predict the rainfall data within 1 h in the future and draw 10 frames of images, and each frame represents the rainfall map within 6 min.

3.2 Evaluation methodology

As shown in Table 2, the frequencies of different rainfall levels are highly imbalanced, and we adopted the same strategy as in [34] to solve this problem by using a weighted loss function. We assign weight $w(x)$ to each pixel according to its rainfall intensity x :

$$w(x) = \begin{cases} 1, & x < 0.1 \\ 2, & 0.1 \leq x < 1 \\ 5, & 1 \leq x < 10 \\ 10, & 10 \leq x < 20 \\ 20, & 20 \leq x \end{cases} \quad (8)$$

and then, we construct our loss function as follows:

$$l(X, \hat{X}) = \frac{1}{2N} \sum_{n=1}^N \sum_{i=1}^{64} \sum_{j=1}^{64} w_{n,i,j} \left((X_{n,i,j} - \hat{X}_{n,i,j})^2 + |X_{n,i,j} - \hat{X}_{n,i,j}| \right) \quad (9)$$

where N is the total number of frames and $w_{n,i,j}$ is the weight corresponding to the (i,j) th pixel in the n th frame. The above loss function combines L1 loss and L2 loss and introduces weights, which effectively alleviates the influence of outliers in the data on the model and the problem of imbalanced distribution of rainfall intensity values.

The prediction model is evaluated by four indicators commonly used in the meteorological community: Critical Success Index (CSI), False Alarm Rate (FAR), Probability of Detection (POD) and Heidke Skill Score (HSS) [45]. CSI, FAR, POD and HSS are defined as

$$\begin{aligned}
 \text{CSI} &= \frac{TP}{TP + TN + FP} \\
 \text{FAR} &= \frac{FP}{TP + TN} \\
 \text{POD} &= \frac{TP}{TP + FP} \\
 \text{HSS} &= \frac{2 * (TP \times TN - FN \times FP)}{(TP + FN)(FN + TN) + (TP + FP) \times (FP + TN)} \quad (10)
 \end{aligned}$$

where TP (prediction = 1, groundtruth = 1), FP (prediction = 1, groundtruth = 0), FN (prediction = 0, groundtruth = 1) and TN (prediction = 0, groundtruth = 0) are calculated by converting the prediction and groundtruth into a 0/1 matrix with a specific rainfall intensity value as a threshold. The higher CSI, POD and HSS and the lower FAR indicate a better prediction result.

3.3 Ablation study

To show the improvement in the proposed 3D-SA-LSTM cell, we use 3D-SA-LSTM to replace the original memory cell in some models and performed an ablation study on the Moving MNIST dataset. All models predict the next 10 frames based on 10 previous frames; we follow the experiment settings and hyper-parameters of ConvLSTM and PredRNN for a fair comparison. The qualitative comparison of predicted video sequences is shown in Fig. 6; it can be seen that the baseline model with 3D-SA-LSTM is significantly better than the original method. Though the predicted trajectories are relatively accurate, the prediction results of the baseline model gradually become blurred, especially at 17–20 frames. Furthermore, digit ‘5’ loses its tail pixels and the details of the upper right corner of the digit ‘8’ are oversimplified. By contrast, the results of the baseline model introduced with 3D-SA-LSTM are not only clear enough but also more accurate for long-term motion prediction. We use mean square error (MSE) as the evaluation indicator of the model; after embedding 3D-SA-LSTM, the sequence MSE in



Fig. 6 Prediction examples on the moving MNIST test set

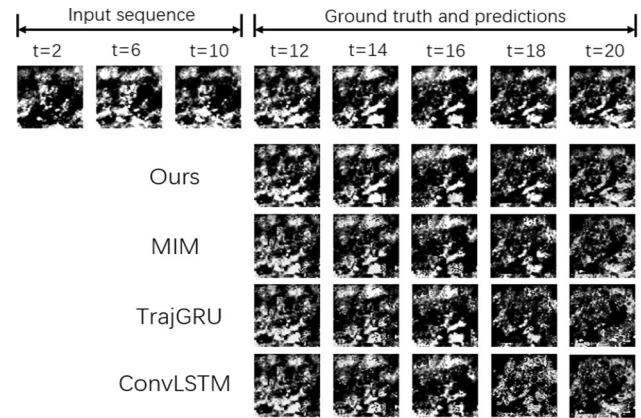


Fig. 7 Prediction examples of the next-hour predictions of the rainfall dataset, where the pixel values indicate higher rainfall intensity value

ConvLSTM and PredRNN is reduced by 11% and 4%, respectively.

3.4 Results analysis and discussion

The prediction results of the four different models are shown in Fig. 7. We can see that the evolution of rainfall distribution is a highly fast and complicated process, and the intensity and region of rainfall are changing all the time. We can find that the prediction results of the ConvLSTM are not very good, which is mainly due to the insufficient modeling ability of the model, and it is difficult for the convolution kernel with fixed connection to capture the long-range spatial dependencies. TrajGRU and MIM significantly deviate from the observed data in the longer-term forecast, especially in the overall movement trend. At the same time, it can be seen that two-stream ConvLSTM is superior to other algorithms, both in detail and in the overall movement trend, which is mainly due to two reasons. First, the introduction of 3D convolution and self-attention mechanism can help our model better capture the spatiotemporal dependencies and learn the details of the rainfall process. Second, the collaborative learning of the two sub-networks helps to improve the performance of model prediction; the long-term sub-network can better

Table 3 Comparison of CSI skill scores of other models. ‘CSI- τ ’ represents the skill scores at τ mm/h rainfall threshold

Model	CSI-0.1	CSI-5	CSI-10	CSI-20
OURS	0.778	0.714	0.595	0.371
MIM	0.734	0.705	0.589	0.306
TrajGRU	0.648	0.679	0.538	0.345
ConvLSTM	0.689	0.652	0.545	0.316

Table 4 Comparison of FAR skill scores of other models. 'FAR- τ ' represents the skill scores at τ mm/h rainfall threshold

Model	FAR -0.1	FAR -5	FAR -10	FAR -20
OURS	0.160	0.101	0.172	0.393
MIM	0.182	0.115	0.185	0.441
TrajGRU	0.297	0.146	0.241	0.419
ConvLSTM	0.209	0.153	0.233	0.472

Table 5 Comparison of POD skill scores of other models. 'POD- τ ' represents the skill scores at τ mm/h rainfall threshold

Model	POD-0.1	POD-5	POD-10	POD-20
OURS	0.913	0.776	0.679	0.488
MIM	0.878	0.775	0.681	0.403
TrajGRU	0.894	0.768	0.648	0.461
ConvLSTM	0.843	0.739	0.654	0.441

Table 6 Comparison of HSS skill scores of other models. 'HSS- τ ' represents the skill scores at τ mm/h rainfall threshold

Model	HSS-0.1	HSS-5	HSS-10	HSS-20
OURS	0.695	0.696	0.602	0.404
MIM	0.628	0.682	0.592	0.322
TrajGRU	0.417	0.644	0.522	0.369
ConvLSTM	0.558	0.614	0.532	0.321

capture the evolution of the movement trend by learning from long-term data.

The overall evaluation results are shown in Tables 3, 4, 56; we choose four thresholds to evaluate the skill scores of the model under different rainfall intensities. Among many indicators, our model has achieved the best results. Only in POD-10, our experimental results are slightly worse than MIM, but MIM will trigger more false alarms. As the threshold increases, the prediction performance will become worse, which is caused by the characteristics of the data distribution, i.e., the heavy rainfall data only occupy a small part of the rainfall dataset.

4 Conclusions

In this paper, we propose a new deep learning model called two-stream convolutional LSTM for precipitation nowcasting. This model uses two sub-network collaboratively to capture the heterogeneity in rainfall data, which can

adequately model the changes in rainfall. Besides, we construct a new memory cell that employs 3D convolutions and self-attention mechanism to capture long-range dependencies both in spatial and in temporal dimensions. We evaluated our networks on a real-world large-scale rainfall dataset, and the experimental results show that our model achieves the state-of-the-art results. In Fig. 7 and Tables 3, 4, 5, 6, we can clearly find that our proposed model has a better performance than other models, which is mainly due to two reasons. On the one hand, the introduction of 3D convolution and self-attention mechanism improves the ability of model to capture spatiotemporal features and can better learn the details of the rainfall process. On the other hand, the collaborative learning of the two sub-networks helps to improve the performance of model prediction; the long-term sub-network can better capture the evolution of the movement trend by learning from long-term data. For future work, we will explore how to solve the problem that the accuracy of prediction decreases over time. One idea is to use stochastic neural networks to model uncertainty to improve prediction performance. In addition, we will also try to perform convolution calculations in the frequency domain to speed up the training process.

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s00521-021-06877-9>.

Author contributions All authors contributed to the study conception and design. Material preparation, data collection and analysis were performed by Suting Chen and Xin Xu. The first draft of the manuscript was written by Xin Xu, and all authors commented on previous versions of the manuscript. All authors read and approved the final manuscript.

Funding This work was supported by the National Natural Science Foundation of China (Nos. 61906097) and a project funded by the Priority Academic Program Development of Jiangsu Higher Education Institutions (PAPD).

Declarations

Conflicts of interest The authors declare that they have no conflict of interests.

References

1. Yuehong S, Wanchang Z, Yonghe L (2009) Application of back-propagation neural network in precipitation estimation with doppler radar. *Plateau Meteor* 28(4):846–853
2. Heuvelink D, Berenguer M, Brauer CC et al (2020) Hydrological application of radar rainfall nowcasting in the Netherlands. *Environ Int* 136:105431. <https://doi.org/10.1016/j.envint.2019.105431>
3. Sun J, Xue M, Wilson JW et al (2014) Use of NWP for nowcasting convective precipitation: Recent progress and challenges.

- Bull Am Meteor Soc 95(3):409–426. <https://doi.org/10.1175/BAMS-D-11-00263.1>
4. Wang G, Wong W, Liu L et al (2013) Application of multi-scale tracking radar echoes scheme in quantitative precipitation nowcasting. *Adv Atmos Sci* 30(2):448–460. <https://doi.org/10.1175/BAMS-D-11-00263.1>
5. Li H, Wang X, Wu S et al (2021) A new method for determining an optimal diurnal threshold of GNSS precipitable water vapor for precipitation forecasting. *Remote Sens* 13(7):1390. <https://doi.org/10.3390/rs13071390>
6. Chen M, Bica B, Tüchler L et al (2017) Statistically extrapolated nowcasting of summertime precipitation over the Eastern Alps. *Adv Atmos Sci* 34:925–938. <https://doi.org/10.1007/s00376-017-6185-4>
7. LeCun Y, Bengio Y, Hinton G (2015) Deep learning. *Nature* 521(7553):436–444. <https://doi.org/10.1038/nature14539>
8. Shi X, Chen Z, Wang H, Yeung DY, Wong WK, Woo WC (2015) Convolutional LSTM network: a machine learning approach for precipitation nowcasting. In: *Advances in neural information processing systems*, p 802–810.
9. Kong Y, Fu Y (2018) Human action recognition and prediction: A survey. [arXiv:1806.11230](https://arxiv.org/abs/1806.11230)
10. Sahin C, Garcia-Hernando G, Sock J et al (2020) A review on object pose recovery: from 3d bounding box detectors to full 6d pose estimators. *Image Vis Comput* 96:103898. <https://doi.org/10.1016/j.imavis.2020.103898>
11. Sun H, Tang M, Peng W et al (2021) Interval prediction of short-term building electrical load via a novel multi-objective optimized distributed fuzzy model. *Neural Comput & Applic*. <https://doi.org/10.1007/s00521-021-06162-9>
12. Boogaard MV, Pickkers P, Hoeven HV et al (2010) PREDICT, Prediction of Delirium in ICU Patients: development and validation of a prediction model. *Crit Care* 14:P498. <https://doi.org/10.1186/cc8730>
13. Chen Z, Xiao X, Li C et al (2016) Erratum to: Real-time transient stability status prediction using cost-sensitive extreme learning machine. *Neural Comput & Applic* 27:333. <https://doi.org/10.1007/s00521-015-1926-8>
14. Baehr J, Fröhlich K, Botzet M et al (2015) The prediction of surface temperature in the new seasonal prediction system based on the MPI-ESM coupled climate model. *Clim Dyn* 44(9–10):2723–2735. <https://doi.org/10.1007/s00382-014-2399-7>
15. Kosek W, McCarthy DD, Luzum BJ (2001) El Niño impact on polar motion prediction errors. *Stud Geophys Geod* 45(4):347–361. <https://doi.org/10.1023/A:1022073503034>
16. Chiu PC, Selamat A, Krejcar O et al (2021) Imputation of rainfall data using the sine cosine function fitting neural network. *Int J Interact Multimed and Art Intellig*. <https://doi.org/10.9781/ijimai.2021.08.013>
17. Sutskever I, Vinyals O, Le Q V (2014) Sequence to sequence learning with neural networks. [arXiv:1409.3215](https://arxiv.org/abs/1409.3215).
18. Schmidhuber J (1992) Learning complex, extended sequences using the principle of history compression. *Neural Comput* 4(2):234–242. <https://doi.org/10.1162/neco.1997.9.8.1735>
19. Cho, K., van Merriënboer, B., Gulcehre, C., et al. (2014). Learning phrase representations using RNN encoder-decoder for statistical machine translation. [arXiv:1406.1078](https://arxiv.org/abs/1406.1078)
20. Srivastava N, Mansimov E, Salakhutdinov R (2015) Unsupervised learning of video representations using LSTMS. In: *International conference on machine learning*, p 843–852
21. Ranzato MA, Huang FJ, Boureau YL et al (2007) Unsupervised learning of invariant feature hierarchies with applications to object recognition[C]. *IEEE Conf Comput Vis Pattern Recogn* 2007:1–8
22. Finn C, Goodfellow I, Levine S (2016) Unsupervised learning for physical interaction through video prediction. [arXiv:1605.07157](https://arxiv.org/abs/1605.07157).
23. Kalchbrenner N, van den Oord A, Simonyan K, Danihelka I, Vinyals O, Graves A, Kavukcuoglu K (2017) Video pixel networks. In: *Proceedings of the international conference on machine learning*. p 1771–1779
24. Vondrick C, Pirsaviash H, Torralba A (2016) Generating videos with scene dynamics. [arXiv:1609.02612](https://arxiv.org/abs/1609.02612)
25. Mathieu M, Couprie C, LeCun Y (2015) Deep multi-scale video prediction beyond mean square error. [arXiv:1511.05440](https://arxiv.org/abs/1511.05440)
26. Goodfellow IJ, Pouget-Abadie J, Mirza M, Xu B, Warde-Farley D, Ozair S, Courville A, Bengio Y (2014) Generative adversarial nets. [arXiv:1406.2661v1](https://arxiv.org/abs/1406.2661v1)
27. Tran D, Bourdev L, Fergus R, Torresani L, Paluri M (2015) Learning spatiotemporal features with 3d convolutional networks. In: *Proceedings of the IEEE International conference on computer vision*, p 4489–4497. <https://doi.org/10.1109/ICCV.2015.510>.
28. Ji S, Xu W, Yang M, Yu K (2013) 3D convolutional neural networks for human action recognition. *IEEE Trans Pattern Anal Mach Intell* 35(1):221–231. <https://doi.org/10.1109/TPAMI.2012.59>
29. Wang X, Girshick R, Gupta A, He K (2018) Non-local neural networks. In: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, p 7794–7803. <https://doi.org/10.1109/CVPR.2018.00813>
30. Vaswani A, Shazeer N, Parmar N et al (2017) Attention is all you need. [arXiv:1706.03762](https://arxiv.org/abs/1706.03762)
31. Huddar MG, Sannakki SS, Rajpurohit VS (2021) Attention-based multi-modal sentiment analysis and emotion detection in conversation using RNN. *Int J Inter Multimed and Art Intellig*. <https://doi.org/10.9781/ijimai.2020.07.004>
32. Klein B, Wolf L, Afek Y (2015) A dynamic convolutional layer for short range weather prediction. In: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, p 4840–4848. <https://doi.org/10.1109/CVPR.2015.7299117>
33. Woo W, Wong W (2017) Operational application of optical flow techniques to radar-based rainfall nowcasting. *Atmos* 8:48. <https://doi.org/10.3390/ATMOS8030048>
34. Shi X, Gao Z, Lausen L et al (2017) Deep learning for precipitation nowcasting: a benchmark and a new model. [arXiv:1706.03458](https://arxiv.org/abs/1706.03458)
35. Wang Y, Long M, Wang J, Gao Z, Yu P S (2017) PredRNN: recurrent neural networks for predictive learning using spatiotemporal LSTMs. In: *Proceedings of the Neural Information Processing Systems*, p 879–888.
36. Wang Y, Jiang L, Yang M H, Li L J, Long M, Li F (2019) Eidetic 3d lstm: A model for video prediction and beyond. In: *Proceedings of the International Conference on Learning Representations*
37. Wang Y, Zhang J, Zhu H, Long M, Wang J, Yu P S (2019) Memory in memory: a predictive neural network for learning higher-order non-stationarity from spatiotemporal dynamics. In: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, p 9154–9162. <https://doi.org/10.1109/CVPR.2019.00937>.
38. Guen, V L, Thome N (2020) Disentangling physical dynamics from unknown factors for unsupervised video prediction. In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, p. 11474–11484. <https://doi.org/10.1109/CVPR42600.2020.01149>
39. Luo W, Li Y, Urtasun R, Zemel R (2016) Understanding the effective receptive field in deep convolutional neural networks. In: *Proceedings of the 30th International Conference on Neural Information Processing Systems (NIPS)*, p 4905–4913
40. Hu J, Shen L, Sun G (2018) Squeeze-and-excitation networks. In: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, p 7132–7141. <https://doi.org/10.1109/TPAMI.2019.2913372>

41. Simonyan K, Zisserman A (2014) Two-stream convolutional networks for action recognition in videos. [arXiv:1406.2199](#)
42. He K, Zhang X, Ren S et al (2016) Deep residual learning for image recognition, In: Proceedings of the 2016 IEEE conference on computer vision and pattern recognition, p 770–778. <https://doi.org/10.1109/cvpr.2016.90>
43. Kingma D P, Ba J (2014) Adam: A method for stochastic optimization. [arXiv:1412.6980](#)
44. Ba J L, Kiros J R, Hinton G E (2016) Layer normalization. [arXiv:1607.06450](#)
45. Hogan RJ, Ferro CAT, Jolliffe IT et al (2010) Equitability revisited: why the “equitable threat score” is not equitable. *Weather Forecast* 25:710–726. <https://doi.org/10.1175/2009WAF2222350.1>

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.